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If $s > 0$ and $\sqrt{\frac{r}{s}} = s$, what is r in terms of s?

A. $\frac{1}{s}$

B. $\sqrt{s}$

C. $s\sqrt{s}$

D. s^3

E. $s^3 - s$

Algebra Equations

Solve the equation for r as follows:

$$\sqrt{\frac{r}{s} = s}$$

$$\frac{r}{s} = s^2 \quad \text{square both sides of the equation}$$

$$r = s^3 \quad \text{multiply both sides by } s$$

The correct answer is D.

•

If $3 < x < 100$, for how many values of x is $\frac{x}{3}$ the square of a prime number?

A. Two

B. Three

C. Four

D. Five

E. Nine

Arithmetic Properties of Numbers

If $\frac{x}{3}$ is the square of a prime number, then possible values of $\frac{x}{3}$ are 2^2, 3^2, 5^2, 7^2,… Therefore, possible values of x are $3 \times 2^2 = 12$, $3 \times 3^2 = 27$, $3 \times 5^2 = 75$, $3 \times 7^2 = 147$, …. Since only three of these values, namely 12, 27, and 75, are between 3 and 100, there are three values of x such that $\frac{x}{3}$ is the square of a prime number.

The correct answer is B.

•

A researcher plans to identify each participant in a certain medical experiment with a code consisting of either a single letter or a pair of distinct letters written in alphabetical order. What is the least number of letters that can be used if there are 12 participants, and each participant is to receive a different code?

A. 4

B. 5

C. 6

D. 7

E. 8

Arithmetic Elementary Combinatorics

None of the essential aspects of the problem is affected if the letters are restricted to be the first n letters of the

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None of the essential aspects of the problem is affected if the letters are restricted to be the first n letters of the alphabet, for various positive integers n. With the 3 letters a, b, and c, there are 6 codes: a, b, c, ab, ac, and bc. With the 4 letters a, b, c, and d, there are 10 codes: a, b, c, d, ab, ac, ad, bc, bd, and cd. Clearly, more than 12 codes are possible with 5 or more letters, so the least number of letters that can be used is 5.

The correct answer is B.

•

An object thrown directly upward is at a height of h feet after t seconds, where $h = -16(t-3)^2 + 150$. At what height, in feet, is the object 2 seconds after it reaches its maximum height?

A. 6

B. 86

C. 134

D. 150

E. 166

Algebra Applied Problems

Since $(t-3)^2$ is positive when $t \neq 3$ and zero when $t = 3$, it follows that the *minimum* value of $(t-3)^2$ occurs when $t = 3$. Therefore, the *maximum* value of $-16(t-3)^2$, and also the maximum value of $-16(t-3)^2 + 150$, occurs when $t = 3$. Hence, the height 2 seconds after the maximum height is the value of h when $t = 5$, or $-16(5-3)^2 + 150 = 86$.

The correct answer is B.

•

Which of the following is equivalent to the pair of inequalities $x + 6 > 10$ and $x - 3 \leq 5$?

A. $2 \leq x < 16$

B. $2 \leq x < 4$

C. $2 < x \leq 8$

D. $4 < x \leq 8$

E. $4 \leq x < 16$

Algebra Inequalities

Solve the inequalities separately and combine the results.

$$\begin{aligned} x + 6 &> 10 \\ x &> 4 \\ x - 3 &\leq 5 \\ x &\leq 8 \end{aligned}$$

Since $x > 4$, then $4 < x$. Combining $4 < x$ and $x \leq 8$ gives $4 < x \leq 8$.

The correct answer is D.

•

David has d books, which is 3 times as many as Jeff and $\frac{1}{2}$ as many as Paula. How many books do the three of them have altogether, in terms of d?

A. $\frac{5}{6}d$

B. $\frac{7}{3}d$

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C. $\frac{10}{3}d$

D. $\frac{7}{2}d$

E. $\frac{9}{2}d$

Algebra Applied Problems; Simultaneous Equations

Let J be the number of books that Jeff has, and let P be the number of books Paula has. Then the given information about David's books can be expressed as $d = 3J$ and $d = \frac{1}{2}P$. Solving these two equations for J and P gives $\frac{d}{3} = J$ and $2d = P$. Thus, $d + J + P = d + \frac{d}{3} + 2d = 3\frac{1}{3}d = \frac{10}{3}d$.

The correct answer is C.

•

There are 8 teams in a certain league and each team plays each of the other teams exactly once. If each game is played by 2 teams, what is the total number of games played?

A. 15

B. 16

C. 28

D. 56

E. 64

Arithmetic Operations on Rational Numbers

Since no team needs to play itself, each team needs to play 7 other teams. In addition, each game needs to be counted only once, rather than once for each team that plays that game. Since two teams play each game, $\frac{8 \times 7}{2} = 28$ games are needed.

The correct answer is C.

•

At his regular hourly rate, Don had estimated the labor cost of a repair job as $336 and he was paid that amount. However, the job took 4 hours longer than he had estimated and, consequently, he earned $2 per hour less than his regular hourly rate. What was the time Don had estimated for the job, in hours?

A. 28

B. 24

C. 16

D. 14

E. 12

Algebra Second-Degree Equations

For ease of calculation later, note that 336 = 6 × 7 × 8.

To avoid fairly extensive algebraic manipulation, consider starting with the answer choices by checking to see which one satisfies the conditions of the problem.

Estimated		Actual	
Number of hours	Per hour pay†	Number of hours	Per hour pay
A 28	$\frac{\$336}{28} = \$12^*$	32	$\frac{\$336}{32} \neq \$10^{**}$
	[illegible]		[illegible]

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B 24	$\frac{\$336}{24} = \$14$	28	$\frac{\$336}{28} = \$12$
C 16	$\frac{\$336}{16} = \$21$	20	$\frac{\$336}{20} \neq \$19$
D 14	$\frac{\$336}{14} = \$24$	18	$\frac{\$336}{18} \neq \$22$
E 12	$\frac{\$336}{12} = \$28$	16	$\frac{\$336}{16} \neq \$26$

† Note that Don used his regular hourly rate to estimate the labor cost of the repair job, so if he estimated $336 for his labor for a t-hour job, then his regular hourly rate was $\frac{\$336}{t}$.

*Cancel wherever possible $\frac{336}{28} = \frac{6 \times \cancel{7} \times \cancel{8}^{2}}{\cancel{7} \times \cancel{4}} = 6 \times 2 = 12$

** This would need to be $12 – $2 = $10 in order for (A) 28 to be correct.

The correct answer is B.

•

If $\frac{p}{q} < 1$, and p and q are positive integers, which of the following must be greater than 1?

A. $\sqrt{\frac{p}{q}}$

B. $\frac{p}{q^2}$

C. $\frac{p}{2q}$

D. $\frac{q}{p^2}$

E. $\frac{q}{p}$

Arithmetic Properties of Numbers

Since p and q are positive integers, $0 < \frac{p}{q} < 1$.

A. Since $\frac{p}{q} < 1$, then $q > p$. Taking the square root of both sides of the inequality gives $\sqrt{q} > \sqrt{p}$. Then, $\sqrt{\frac{p}{q}} = \frac{\sqrt{p}}{\sqrt{q}}$, so here the denominator will still be larger than the numerator. CANNOT be greater than 1.

B. Squaring the denominator increases the denominator, which decreases the value of the fraction. CANNOT be greater than 1.

C. Multiplying the denominator by 2 increases the denominator, which decreases the value of the fraction. CANNOT be greater than 1.

D. Since $\frac{p}{q} < 1$, then $q > p$. When $p^2 < q$, this expression will be greater than 1, but p^2 need not be less than q. For example, $p = 2$ if and $q = 100$, $\frac{p}{q} = \frac{2}{100}$ and $\frac{q}{p^2} = \frac{100}{2^2} = \frac{100}{4} = 25 > 1$.

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However, if $p = 3$ and $q = 4$, then $\frac{p}{q} = \frac{3}{4}$ and $\frac{q}{p^2} = \frac{4}{3^2} = \frac{4}{9} < 1$. NEED NOT be greater than 1.

E. Again, since $\frac{p}{q} < 1$, then $q > p$. Thus, the reciprocal, $\frac{q}{p}$, always has a value greater than 1 because the numerator will always be a larger positive integer than the denominator. MUST be greater than 1.

The correct answer is E.

•

To mail a package, the rate is x cents for the first pound and y cents for each additional pound, where $x > y$. Two packages weighing 3 pounds and 5 pounds, respectively, can be mailed separately or combined as one package. Which method is cheaper, and how much money is saved?

A. Combined, with a savings of $x - y$ cents

B. Combined, with a savings of $y - x$ cents

C. Combined, with a savings of x cents

D. Separately, with a savings of $x - y$ cents

E. Separately, with a savings of y cents

Algebra Applied Problems

Shipping the two packages separately would cost $1x + 2y$ for the 3-pound package and $1x + 4y$ for the 5-pound package. Shipping them together (as a single 8-pound package) would cost $1x + 7y$. By calculating the sum of the costs for shipping the two packages separately minus the cost for shipping the one combined package, it is possible to determine the difference in cost, as shown.

$((1x + 2y) + (1x + 4y)) - (1x + 7y)$	(cost for 3 lbs + cost for 5 lbs) – cost for 8 lbs
$= (2x + 6y) - (1x + 7y)$	combine like terms
$= 2x + 6y - 1x - 7y$	distribute the negative
$= x - y$	combine like terms

Since $x > y$, this value is positive, which means it costs more to ship two packages separately. Thus, it is cheaper to mail one combined package at a cost savings of $x - y$ cents.

The correct answer is A.

•

If money is invested at r percent interest, compounded annually, the amount of the investment will double in approximately $\frac{70}{r}$ years. If Pat's parents invested $5,000 in a long-term bond that pays 8 percent interest, compounded annually, what will be the approximate total amount of the investment 18 years later, when Pat is ready for college?

A. $20,000

B. $15,000

C. $12,000

D. $10,000

E. $9,000

Algebra Applied Problems

Since the investment will double in $\frac{70}{r} = \frac{70}{8} = 8.75 \approx 9$ years, the value of the investment over 18 years can be approximated by doubling its initial value twice. Therefore, the approximate value will be ($5,000)(2)(2) = $20,000.

The correct answer is A.

•

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•

On a recent trip, Cindy drove her car 290 miles, rounded to the nearest 10 miles, and used 12 gallons of gasoline, rounded to the nearest gallon. The actual number of miles per gallon that Cindy's car got on this trip must have been between

A. $\frac{290}{12.5}$ and $\frac{290}{11.5}$

B. $\frac{295}{12}$ and $\frac{285}{11.5}$

C. $\frac{285}{12}$ and $\frac{295}{12}$

D. $\frac{285}{12.5}$ and $\frac{295}{11.5}$

E. $\frac{295}{12.5}$ and $\frac{285}{11.5}$

Arithmetic Estimation

The lowest number of miles per gallon can be calculated using the lowest possible miles and the highest amount of gasoline. Also, the highest number of miles per gallon can be calculated using the highest possible miles and the lowest amount of gasoline.

Since the miles are rounded to the nearest 10 miles, the number of miles is between 285 and 295. Since the gallons are rounded to the nearest gallon, the number of gallons is between 11.5 and 12.5. Therefore, the lowest number of miles per gallon is $\frac{\text{lowest miles}}{\text{highest gallons}} = \frac{285}{12.5}$ and the highest number of miles per gallon is $\frac{\text{highest miles}}{\text{lowest gallons}} = \frac{295}{11.5}$

The correct answer is D.

•

Which of the following inequalities is an algebraic expression for the shaded part of the number line above?

A. $|x| \leq 3$

B. $|x| \leq 5$

C. $|x-2| \leq 3$

D. $|x-1| \leq 4$

E. $|x+1| \leq 4$

Algebra Inequalities

The number line above shows $-5 \leq x \leq 3$. To turn this into absolute value notation, as all the choices are written, the numbers need to be opposite signs of the same value.

Since the distance between −5 and 3 is 8 (3 – (–5) = 8), that distance needs to be split in half with −4 to one side and 4 to the other. Each of these two values is 1 more than the values in the inequality above, so adding 1 to all terms in the inequality gives $-4 \leq x+1 \leq 4$, which is the same as $|x+1| \leq 4$.

The correct answer is E.

•

In a small snack shop, the average (arithmetic mean) revenue was $400 per day over a 10-day period. During this period, if the average daily revenue was $360 for the first 6 days, what was the average daily revenue for the last 4

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period, if the average daily revenue was $360 for the first 6 days, what was the average daily revenue for the last 4 days?

A. $420

B. $440

C. $450

D. $460

E. $480

Arithmetic; Algebra Statistics; Applied Problems

Let x be the average daily revenue for the last 4 days. Using the formula $\text{average} = \dfrac{\text{sum of values}}{\text{number of values}}$, the information regarding the average revenues for the 10-day and 6-day periods can be expressed as follows and solved for x:

$$\$400 = \frac{6(\$360) + 4x}{10}$$

$$\$4{,}000 = \$2{,}160 + 4x \quad \text{multiply both sides by 10}$$

$$\$1{,}840 = 4x \quad \text{subtract \$2, 160 from both sides}$$

$$\$460 = x \quad \text{divide both sides by 4}$$

The correct answer is D.

•

If y is the smallest positive integer such that 3,150 multiplied by y is the square of an integer, then y must be

A. 2

B. 5

C. 6

D. 7

E. 14

Arithmetic Properties of Numbers

To find the smallest positive integer y such that $3,150y$ is the square of an integer, first find the prime factorization of 3,150 by a method similar to the following:

$$\begin{aligned} 3{,}150 &= 10 \times 315 \\ &= (2 \times 5) \times (3 \times 105) \\ &= 2 \times 5 \times 3 \times (5 \times 21) \\ &= 2 \times 5 \times 3 \times 5 \times (3 \times 7) \\ &= 2 \times 3^2 \times 5^2 \times 7 \end{aligned}$$

To be a perfect square, $3,150y$ must have an even number of each of its prime factors. At a minimum, y must have one factor of 2 and one factor of 7 so that $3,150y$ has two factors of each of the primes 2, 3, 5, and 7. The smallest positive integer value of y is then (2)(7) = 14.

The correct answer is E.

•

If $[x]$ is the greatest integer less than or equal to x, what is the value of [–1.6] + [3.4] + [2.7]?

A. 3

B. 4

C. 5

D. 6

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D. 6

E. 7

Arithmetic Profit and Loss

The greatest integer that is less than or equal to −1.6 is −2. It cannot be −1 because −1 is greater than −1.6. The greatest integer that is less than or equal to 3.4 is 3. It cannot be 4 because 4 is greater than 3.4. The greatest integer that is less than or equal to 2.7 is 2. It cannot be 3 because 3 is greater than 2.7. Therefore, [–1.6] + [3.4] + [2.7] = –2 + 3 + 2 = 3.

The correct answer is A.

•

In the first week of the year, Nancy saved $1. In each of the next 51 weeks, she saved $1 more than she had saved in the previous week. What was the total amount that Nancy saved during the 52 weeks?

A. $1,326

B. $1,352

C. $1,378

D. $2,652

E. $2,756

Arithmetic Operations on Rational Numbers

In dollars, the total amount saved is the sum of 1, (1 + 1), (1 + 1 + 1), and so on, up to and including the amount saved in the 52nd week, which was $52. Therefore, the total amount saved in dollars was 1 + 2 + 3 + … + 50 + 51 + 52. This sum can be easily evaluated by grouping the terms as (1 + 52) + (2 + 51) + (3 + 50) + … + (26 + 27), which results in the number 53 added to itself 26 times. Therefore, the sum is (26)(53) = 1,378.

Alternatively, the formula for the sum of the first n positive integers is $\frac{n(n+1)}{2}$. Therefore, the sum of the first 52 positive integers is $\frac{52(53)}{2} = 26\left(53\right) = 1,378$.

The correct answer is C.

•

In a certain sequence, the term x_n is given by the formula $x_n = 2x_{n-1} - \frac{1}{2}\left(x_{n-2}\right)$ for all $n \geq 2$. If $x_0 = 3$ and $x_1 = 2$, what is the value of x_3?

A. 2.5

B. 3.125

C. 4

D. 5

E. 6.75

Algebra Simplifying Algebraic Expressions

Given the formula $x_n = 2x_{n-1} - \frac{1}{2}\left(x_{n-2}\right)$ with $x_0 = 3$ and $x_1 = 2$, then

$$\begin{aligned} x_2 &= 2x_1 - \frac{1}{2}x_0 \\ &= 2\left(2\right) - \frac{1}{2}\left(3\right) \\ &= \frac{5}{2} \end{aligned}$$

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$$
\begin{aligned}
- & \quad 2 \\
x_2 &= 2x_2 - \frac{1}{2}x_1 \\
&= 2\left(\frac{5}{2}\right) - \frac{1}{2}\left(2\right) \\
&= 5-1 \\
&= 4
\end{aligned}
$$

The correct answer is C.

•

During a trip, Francine traveled x percent of the total distance at an average speed of 40 miles per hour and the rest of the distance at an average speed of 60 miles per hour. In terms of x, what was Francine's average speed for the entire trip?

A. $\frac{180-x}{2}$

B. $\frac{x+60}{4}$

C. $\frac{300-x}{5}$

D. $\frac{600}{115-x}$

E. $\frac{12{,}000}{x+200}$

Algebra Applied Problems

Assume for simplicity that the total distance of Francine's trip is 100 miles. Then the following table gives all of the pertinent information.

Distance	Rate	>
x	40	$\frac{x}{40}$
$100-x$	60	$\frac{100-x}{60}$

The total time for Francine's trip is

$$
\begin{aligned}
\frac{x}{40} + \frac{100-x}{60} &= \frac{3x}{120} + \frac{2\,(100-x)}{120} \\
&= \frac{3x + 2\,(100-x)}{120} \\
&= \frac{3x + 200 - 2x}{120} \\
&= \frac{x+200}{120}
\end{aligned}
$$

Francine's average speed over the entire trip is $\frac{\text{total distance}}{\text{total time}} = \frac{100}{\frac{x+200}{120}} = \frac{12{,}000}{x+200}$.

The correct answer is E.

•

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If $n = (33)^{43} + (43)^{33}$, what is the units digit of n?

A. 0

B. 2

C. 4

D. 6

E. 8

Arithmetic Properties of Numbers

When a polynomial is raised to a power, the last term in the expression is the last term of the polynomial raised to that power. For example, $(a+b)^2 = a^2 + 2ab + b^2$. We can think of 33 as the binomial $10 \cdot 3 + 3$ whose last term is the units digit 3. Thus, the units digit of 33^2 is $3^2 = 9$. Similarly, for 33^3 we have $3^3 = 27$, so the units digit of 33^3 is 7. The patterns for powers of 33 and 43 are shown in the following table.

power of 33	units digit	power of 43	units digit
33^1	3	43^1	3
33^2	9	43^2	9
33^3	7	43^3	7
33^4	1	43^4	1
33^5	3	43^5	3
33^6	9	43^6	9
33^7	7	43^7	7
33^8	1	43^8	1
⋮	⋮	⋮	⋮
33^{32}	1	43^{32}	1
33^{33}	3	43^{33}	3
⋮	⋮		
33^{40}	1		
33^{41}	3		
33^{42}	9		
33^{43}	7		

From the table above, the units digit of 3^{43} is 7 and the units digit of 43^{33} is 3. Since 7 + 3 = 10, the units digit of $33^{43} + 43^{33}$ is 0.

The correct answer is A.

•

Team A and Team B are competing against each other in a game of tug-of-war. Team A, consisting of 3 males and 3 females, decides to line up male, female, male, female, male, female. The lineup that Team A chooses will be one of how many different possible lineups?

A. 9

B. 12

C. 15

D. 36

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D. 36

E. 720

Arithmetic Elementary Combinatorics

Any of the 3 males can be first in the line, and any of the 3 females can be second. Either of the 2 remaining males can be next, followed by either of the 2 remaining females. The last 2 places in the line are filled with the only male left followed by the only female left. By the multiplication principle, there are 3 × 3 × 2 × 2 × 1 × 1 = 36 different lineups possible.

The correct answer is D.

•

If $d = \frac{1}{2^3 \times 5^7}$ is expressed as a terminating decimal, how many nonzero digits will d have?

A. One

B. Two

C. Three

D. Seven

E. Ten

Arithmetic Operations on Rational Numbers

It will be helpful to use the fact that a factor that is an integer power of 10 has no effect on the number of nonzero digits a terminating decimal has.

$$\begin{aligned}\frac{1}{2^3 \times 5^7} &= \frac{1}{2^3 \times 5^3} \times \frac{1}{5^4} \\ &= \left(\frac{1}{2 \times 5}\right)^3 \times \left(\frac{1}{5}\right)^4 \\ &= \left(\frac{1}{10}\right)^3 \times \left(\frac{1}{5}\right)^4 \\ &= 10^{-3} \times (0.2) \\ &= 10^{-3} \times (0.0016) \\ &= 0.0000016\end{aligned}$$

The correct answer is B.

•

For any positive integer n, the sum of the first n positive integers equals $\frac{n(n+1)}{2}$. What is the sum of all the even integers between 99 and 301?

A. 10,100

B. 20,200

C. 22,650

D. 40,200

E. 45,150

Algebra Simplifying Expressions; Arithmetic Computation with Integers

Note: The following approach does not rely on the given formula.

To determine how many integers are to be summed, consider the following:

5 even integers have 10 as their first two digits, namely 100, 102, 104, 106, 108; 5 even integers have 11 as their first two digits, namely 110, 112, 114, 116, 118;

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two digits, namely 110, 112, 114, 116, 118;

⋮

5 even integers have 28 as their first two digits, namely 280, 282, 284, 286, 288;

5 even integers have 29 as their first two digits, namely 290, 292, 294, 296, 298.

The first two digits range from 10 through 29, so there are 29 – 10 + 1 = 20 such pairs of two digits. Thus, there are 20(5) = 100 even integers between 100 and 298. Including 300, there are 101 even integers between 99 and 301.

These even integers are listed below.

1st	2nd	3rd	4th	…	50th	51st	52nd	…	98th	99th	100th	101st
100	102	104	106	…	198	200	202	…	294	296	298	300

To sum, pair the integers

$$\begin{array}{ll} 1^{st} \text{ and } 101^{st} & 100 + 300 = 400 \\ 2^{nd} \text{ and } 100^{th} & 102 + 298 = 400 \\ 3^{rd} \text{ and } 99^{th} & 104 + 296 = 400 \\ 4^{th} \text{ and } 98^{th} & 106 + 294 = 400 \\ \quad\vdots & \\ 50^{th} \text{ and } 52^{nd} & 198 + 202 = 400 \\ 51^{st} & \qquad\qquad = 200 \end{array} \quad \begin{array}{r} \left.\begin{array}{r} \\ \\ \\ \\ \\ \\ \end{array}\right\} 50(400) = 20{,}000 \\ + \quad 200 \\ \hline 20{,}200 \end{array}$$

Alternatively, let S represent the sum. Then,

$S = 100 + 102 + 104 + \ldots + 296 + 298 + 300$ next, rewrite the sum

$S = 300 + 298 + 296 + \ldots + 104 + 102 + 100$ then add the two equations

$2S = 400 + 400 + 400 + \ldots + 400 + 400 + 400$ now divide by 2

$S = 200 + 200 + 200 + \ldots + 200 + 200 + 200$ there are 101 terms in S, so

$S = 101(200) = 20,200.$

The correct answer is B.

•

November 16, 2001, was a Friday. If each of the years 2004, 2008, and 2012 had 366 days, and the remaining years from 2001 through 2014 had 365 days, what day of the week was November 16, 2014?

A. Sunday

B. Monday

C. Tuesday

D. Wednesday

E. Thursday

Arithmetic Operations with Integers

The number of days between November 16, 2001, and November 16, 2014, is 10(365) + 3(366) = 4,748 and, because 4,748 = 7(678) + 2, it follows that 4,748 days is 678 weeks plus 2 days. Then, 678 weeks from Friday, November 16, 2001, is Friday, November 14, 2014, and 2 days from Friday, November 14, 2014, is Sunday, November 16, 2014.

The correct answer is A.

•

How many prime numbers between 1 and 100 are factors of 7,150?

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How many prime numbers between 1 and 100 are factors of 7,150?

A. One

B. Two

C. Three

D. Four

E. Five

Arithmetic Rate

To find the number of prime numbers between 1 and 100 that are factors of 7,150, find the prime factorization of 7,150 using a method similar to the following:

$$\begin{aligned} 7{,}150 &= 10 \times 715 \\ &= (2 \times 5) \times (5 \times 143) \\ &= 2 \times 5 \times 5 \times (11 \times 13) \end{aligned}$$

Thus, 7,150 has four prime factors: 2, 5, 11, and 13.

The correct answer is D.

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A sequence of numbers $a_1, a_2, a_3, \ldots$ is defined as follows: $a_1 = 3$, $a_2 = 5$, and every term in the sequence after a_2 is the product of all terms in the sequence preceding it, e.g., $a_3 = (a_1)(a_2)$ and $a_4 = (a_1)(a_2)(a_3)$. If $a_n = t$ and $n > 2$, what is the value of a_{n+2} in terms of t?

A. $4t$

B. t^2

C. t^3

D. t^4

E. t^8

Algebra Sequences

It is given that $a_n = (a_1)(a_2)\ \ldots\ (a_{n-1})$ and $a_n = t$. Therefore,
$a_{n+1} = (a_1)(a_2)\ \ldots\ (a_{n-1})(a_n) = (a_n)(a_n) = t^2$ and
$a_{n+2} = (a_1)(a_2)\ \ldots\ (a_n)(a_{n+1}) = (a_{n+1})(a_{n+1}) = (t^2)(t^2) = t^4$.

The correct answer is D.

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Last year the price per share of Stock X increased by k percent and the earnings per share of Stock X increased by m percent, where k is greater than m. By what percent did the ratio of price per share to earnings per share increase, in terms of k and m?

A. $\frac{k}{m}\%$

B. $(k-m)\%$

C. $\frac{100\,(k-m)}{100+k}\%$

D. $\frac{100\,(k-m)}{100+m}\%$

E. $\frac{100\,(k-m)}{100+k+m}\%$

Algebra Percents

Although this problem can be solved algebraically, the answer can also be identified by choosing numerical values

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Although this problem can be solved algebraically, the answer can also be identified by choosing numerical values for the variables, calculating the percent increase in the ratio, and comparing the result with the corresponding numerical values of the answer choices. To this end, let the original price per share be \$1,000 and earnings per share be \$200, where $k = 10$ and $m = 5$. Note that $k > m$ for these values.

	Before increase	After increase
price	\$1,000	1.1(\$1,000) = \$1,100
earnings	\$200	1.05(\$200) = \$210
$\frac{\text{price}}{\text{earnings}}$	$\frac{1{,}000}{200} = 5$	$\frac{1{,}100}{210} = \frac{110}{21}$

Therefore, the percent increase in the ratio of price per share to earnings per share is

$$\left(\frac{\frac{110}{21}}{-} \times 100\right)\% = \left(\frac{110 - 105\,(5)}{(21)} \times 100\right)\% = \frac{500}{105}\%$$

Now substitute $k = 10$ and $m = 5$ into the answer choices to determine which answer choice gives $\frac{500}{105}\%$.

A $\frac{10}{5}\%$ NO

B (10 – 5)% NO

C $\frac{100\,(5)}{100 + 10}\% = \frac{500}{110}\%$ NO

D $\frac{100\,(5)}{100 + 5}\% = \frac{500}{105}\%$**YES**

E $\frac{100\,(5)}{100 + 10 + 5}\% = \frac{500}{115}\%$ NO

The correct answer is D.

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Of the 300 subjects who participated in an experiment using virtual-reality therapy to reduce their fear of heights, 40 percent experienced sweaty palms, 30 percent experienced vomiting, and 75 percent experienced dizziness. If all of the subjects experienced at least one of these effects and 35 percent of the subjects experienced exactly two of these effects, how many of the subjects experienced only one of these effects?

A. 105

B. 125

C. 130

D. 180

E. 195

Arithmetic Applied Problems

Let a be the number who experienced only one of the effects, b be the number who experienced exactly two of the effects, and c be the number who experienced all three of the effects. Then $a + b + c = 300$, since each of the 300 participants experienced at least one of the effects. From the given information, $b = 105$ (35% of 300), which gives $a + 105 + c = 300$, or $a + c = 195$ (Equation 1). Also, if the number who experienced sweaty palms (40% of 300, or 120) is added to the number who experienced vomiting (30% of 300, or 90), and this sum is added to the number who experienced dizziness (75% of 300, or 225), then each participant who experienced only one of the effects is counted exactly once, each participant who experienced exactly two of the effects is counted exactly twice, and each participant who experienced all three of the effects is counted exactly 3 times. Therefore, $a + 2b + 3c = 120 + 90 + 225 = 435$. Using $b = 105$, it follows that $a + 2(105) + 3c = 435$ or $a + 3c = 225$ (Equation 2). Then, solving the system defined by Equation 1 and Equation 2

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$a + 3c = 225$ (Equation 2). Then, solving the system defined by Equation 1 and Equation 2,

$\begin{cases} a + c = 195 \\ a + 3c = 225 \end{cases}$	multiply 1st equation by –3
$\begin{cases} -3a - 3c = -585 \\ a + 3c = 225 \end{cases}$	add equations
$-2a = -360$, or $a = 180$	

The correct answer is D.

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If $m^{-1} = -\frac{1}{3}$, then m^{-2} is equal to

A. –9

B. –3

C. $-\frac{1}{9}$

D. $\frac{1}{9}$

E. 9

Arithmetic Negative Exponents

Using rules of exponents, $m^{-2} = m^{-1\times 2} = \left(m^{-1}\right)^2$, and since $m^{-1} = -\frac{1}{3}$, $m^{-2} = \left(-\frac{1}{3}\right)^2 = \frac{1}{9}$.

The correct answer is D.

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